# Dalplex Data Design

Problem 1: Entity Identification and Design Issues

## 1. Summary

This document records the data design for Dalplex, Dalhousie University's largest athletic and recreation facility. It identifies 22 entities from the gathered business rules. The design resolved many-to-many relationships through an associative entity, was reviewed and found two fan traps, two chasm traps and fifteen entities whose data changes over time. Each issue is paired with a proposed solution.

## 2. Background

### 2.1 About Dalplex

Dalplex is Dalhousie's Recreation and Athletic Centre and its largest athletic facility. It sits at 6260 South Street on the Studley campus in Halifax, and its 16,722 square metres of activity space (Dalhousie University, "Dalplex," Campus Maps). The building opened on 19 Oct. 1979 (Dalhousie University Libraries).

The facility is made up of several distinct areas. The fieldhouse holds four full-size basketball courts, five volleyball courts, nine badminton courts, an indoor running track and a climbing wall. The fitness centre has a fitness hall with strength and cardio equipment, four hardwood studios and a dedicated spin studio. Dalplex also has an eight-lane, 50-metre indoor pool, plus five squash courts and one racquetball court on its lowest level (Dalhousie University, "Dalplex Facilities").

Access is offered through memberships and passes. Besides serving as the student recreation facility, Dalplex sells memberships to faculty and staff, alumni and the community, and it runs both no-fee and paid programs (Dalhousie University, "Dalplex," Athletics & Recreation). The business rules gathered for this assignment cover these areas, together with bookings, rentals and staff.

### 2.2 Entities Identified

Table 1 lists the entities chosen from those business rules, grouped by area. The type column shows whether an entity is a core entity, a lookup, or an associative entity that resolves a many-to-many relationship.

**Table 1.** Entities identified and the reason each was chosen

| Entity | Reason |
|---|---|
| **FACILITY** | Fieldhouse, Golf Room, Fitness Center, Aquatic Center and similar areas are the top-level places. Each has its own area, occupancy and contact details. |

| Entity | Reason |
|---|---|
| **SPACE** | Courts, tracks and rooms share an identifying number, a schedule, rentability and a booking or drop-in mode, so they are modelled as one supertype. |
| **OPERATING_HOURS** | A facility has many time ranges (by day or season), which is a repeating group. |
| **SCHEDULE_SLOT** | The schedule for a space. It lets a space be programmed, bookable or drop-in at a given time. |
| **BOOKING** | A rental or booking made by a patron for one or more slots. |
| **PROGRAM_CATEGORY** | Covers First Aid & CPR, Fitness & wellness, swim lessons, clubs, intramurals and varsity games. A lookup avoids repeating category names as text. |
| **PROGRAM** | A paid or no-fee offering with its own rules and price. |
| **PROGRAM_SESSION** | A program meets many times, and each session takes a slot and has an instructor. |
| **REGISTRATION** | Resolves the PATRON M:N PROGRAM relationship. |
| **STAFF** | Dalplex has staff, and they lead sessions. |
| **PATRON** | One person record for anyone who buys, registers or books. This avoids duplicating people across members, pass holders and renters. |
| **MEMBERSHIP_GROUP** | Full-time students, Faculty, Alumni and so on. Rates, discounts and access hang off the group. |
| **MEMBERSHIP_RATE** | Each group has several prices (term, 1-year individual or family, add-ons). |
| **DISCOUNT** | Optional discounts that can apply to a group. |
| **MEMBERSHIP** | The purchase of a rate by a person, which is what actually grants access. |
| **MEMBERSHIP_HOLDER** | Resolves MEMBERSHIP M:N PATRON, since one membership can cover a spouse or dependants. |
| **PASS_RATE** | The price list for single-day and multi-day passes by age. |
| **PASS** | An issued pass, with an expiry date and a photo ID check. |
| **ADDITIONAL_SERVICE** | Optional extras offered by Dalplex. |
| **PASS_SERVICE** | Resolves PASS M:N ADDITIONAL_SERVICE. |
| **MEMBERSHIP_FACILITY_ACCESS** | Resolves MEMBERSHIP_GROUP M:N FACILITY ("membership grants access to facilities"). |
| **PASS_FACILITY_ACCESS** | Resolves PASS_RATE M:N FACILITY. |

*COURT, TRACK and ROOM are subtypes of SPACE and are not listed separately.*

## 3. Design Issues

This section covers the design issues found in the model: fan traps, chasm traps and time-variant data. Each is described in the same way: where it occurs, what goes wrong, and how it is solved.

### 3.1 Fan Traps

A fan trap occurs when one entity fans out 1:N to two or more other entities. The path between those branches is then ambiguous.

**Table 2.** Fan traps and solutions

| Where it occurs | Problem | Solution |
|---|---|---|
| MEMBERSHIP_GROUP fans out to MEMBERSHIP_RATE, DISCOUNT and MEMBERSHIP_FACILITY_ACCESS. | The model cannot show which discount applies to which rate. | Relate DISCOUNT to MEMBERSHIP_RATE (1:1) so each discount is tied to the rate it modifies. Also relate MEMBERSHIP to MEMBERSHIP_GROUP (1:1) so every purchase traces back to its group. |
| PASS_RATE fans out to PASS and PASS_FACILITY_ACCESS. | Counting passes repeats each pass once for every facility it can access. | Keep PASS_RATE 1:M PASS and attach PASS_FACILITY_ACCESS to PASS (PASS 1:M PASS_FACILITY_ACCESS) instead of PASS_RATE. The entities then form a single chain with no fan-out. |

### 3.2 Chasm Traps

A chasm trap occurs when a path exists in the model but breaks for some rows because participation in a relationship is optional.

**Table 3.** Chasm traps and solutions

| Path | Problem | Solution |
|---|---|---|
| SPACE → SCHEDULE_SLOT → PROGRAM_SESSION | A slot may have no session, so facilities cannot be reliably linked to programs. | Add a direct PROGRAM-to-FACILITY relationship so a program's venue no longer depends on a slot having a session. |
| STAFF → PROGRAM_SESSION | Staff may not lead anything, and sessions may have no instructor, so the path is incomplete in both directions. | Make the instructor mandatory on PROGRAM_SESSION so every session has a staff member leading it. Staff who lead no sessions can stay optional, since that hides no sessions. |

### 3.3 Time-Variant Data

Time-variant data is data whose values change over time while past values still matter. If an update overwrites the old value, past prices, access rights and schedules cannot be reproduced. Fifteen entities are affected, as shown in Table 4.

**Table 4.** Time-variant entities

| Entity | What changes | Why history is needed |
|---|---|---|
| **PASS_RATE** | Price and age bands change over time. | Shows what a patron was charged when the pass was bought. |
| **MEMBERSHIP_RATE** | Prices, terms and add-on prices change by term or year. | Shows what a patron was charged when the membership was bought. |
| **DISCOUNT** | Discount values and which groups receive them change. | Shows which discount applied to a past purchase. |
| **ADDITIONAL_SERVICE** | Service offerings and prices change. | Shows what extras cost when they were sold. |
| **MEMBERSHIP** | Term dates and status (active, expired, cancelled) change. | Shows who held a membership and when. |
| **MEMBERSHIP_HOLDER** | Spouses and dependents are added or removed during a term. | Shows who was covered on a given date. |
| **PASS** | Expiry and status change. | Shows whether a pass was valid on a given date. |
| **REGISTRATION** | Status changes (registered, withdrawn, completed). | Supports refunds and attendance records. |
| **BOOKING** | Dates, times and status change, including cancellations. | Settles booking disputes and fee questions. |
| **PASS_FACILITY_ACCESS** | Facilities a pass grants access to change. | Shows a pass at the time of use. |
| **MEMBERSHIP_FACILITY_ACCESS** | Facilities a membership grants access to change. | Shows what a membership allowed at the time of use. |
| **PROGRAM** | Offerings, fees and schedules change each term. | Keeps past offerings reportable. |
| **PROGRAM_SESSION** | Dates, instructors and cancellations change. | Shows who led a session and whether it ran. |
| **SPACE** | Availability and booking or drop-in mode change. | Shows how a space could be used at a given time. |

| Entity | What changes | Why history is needed |
|---|---|---|
| **STAFF** | Roles and assignments change. | Shows who held which role on a given date. |

# 4. Logical Design

Each entity in the final ERD becomes a table, and each 1:N relationship is stored as a foreign key on the "many" side. Primary keys are underlined and foreign keys are marked (FK). Derived attributes (Age, Duration, SpotsRemaining, IsActive and IsExpired) are calculated when needed and are not stored.

## 4.1 Tables, Dependencies and Prime Attributes

Every table has a single-column primary key, and the four associative tables also have a composite candidate key made of their two foreign keys.

**Table 5.** Tables, functional dependencies and prime attributes (before normalization)

| Table | Columns | Functional dependencies | Prime attributes |
|---|---|---|---|
| **FACILITY** | FacilityID, FType, Area, MaxOccupancy, Address, Contact | FacilityID → all other columns | FacilityID |
| **SPACE** | SpaceID, SpaceNo, SpaceType, UsageMode, IsRentable, FacilityID (FK) | SpaceID → all other columns | SpaceID |
| **OPERATING_HOURS** | HoursID, DayOfWeek, OpenTime, CloseTime, FacilityID (FK) | HoursID → all other columns | HoursID |
| **SCHEDULE_SLOT** | SlotID, StartTime, EndTime, SlotType, SpaceID (FK), BookingID (FK) | SlotID → all other columns | SlotID |
| **BOOKING** | BookingID, BookingType, Fee, BookingDate, PatronID (FK) | BookingID → all other columns | BookingID |
| **PROGRAM_CATEGORY** | CategoryID, CategoryName | CategoryID → all other columns | CategoryID |
| **PROGRAM** | ProgramID, ProgramName, IsPaid, Price, MembersOnly, Capacity, CategoryID (FK), FacilityID (FK) | ProgramID → all other columns | ProgramID |

| Table | Columns | Functional dependencies | Prime attributes |
|---|---|---|---|
| **PROGRAM_SESSION** | SessionID, SessionDate, StartTime, EndTime, SessionStatus, ProgramID (FK), StaffID (FK) | SessionID → all other columns | SessionID |
| **REGISTRATION** | RegID, RegDate, FeePaid, RegStatus, PatronID (FK), ProgramID (FK) | RegID → all other columns. (PatronID, ProgramID) → RegID | RegID, PatronID, ProgramID |
| **STAFF** | StaffID, StaffName, Role, HireDate, FacilityID (FK) | StaffID → all other columns | StaffID |
| **PATRON** | PatronID, PatronName, DOB, Email | PatronID → all other columns | PatronID |
| **MEMBERSHIP_GROUP** | GroupID, GroupName | GroupID → all other columns | GroupID |
| **MEMBERSHIP_RATE** | RateID, Price, Term, TermStart, TermEnd, SpouseAddOn, DependantAddOn, GroupID (FK), DiscountID (FK) | RateID → all other columns. Term → TermStart, TermEnd | RateID |
| **DISCOUNT** | DiscountID, DiscountName, DiscountPct, GroupID (FK) | DiscountID → all other columns | DiscountID |
| **MEMBERSHIP** | MembershipID, StartDate, EndDate, PricePaid, RateID (FK), GroupID (FK) | MembershipID → all other columns. RateID → GroupID | MembershipID |
| **MEMBERSHIP_HOLDER** | HolderID, HolderRole, MembershipID (FK), PatronID (FK) | HolderID → all other columns. (MembershipID, PatronID) → HolderRole | HolderID, MembershipID, PatronID |
| **PASS_RATE** | PassRateID, PassKind, AgeBand, Price, Days | PassRateID → all other columns | PassRateID |
| **PASS** | PassID, IssueDate, ExpiryDate, PhotoIDVerified, PassRateID (FK), PatronID (FK) | PassID → all other columns | PassID |
| **ADDITIONAL_SERVICE** | ServiceID, ServiceName, Price, FacilityID (FK) | ServiceID → all other columns | ServiceID |

| Table | Columns | Functional dependencies | Prime attributes |
|---|---|---|---|
| **MEMBERSHIP_FACILITY_ACCESS** | MemAccessID, Conditions, GroupID (FK), FacilityID (FK) | MemAccessID → all other columns. (GroupID, FacilityID) → Conditions | MemAccessID, GroupID, FacilityID |
| **PASS_FACILITY_ACCESS** | PassAccessID, Conditions, PassID (FK), FacilityID (FK) | PassAccessID → all other columns. (PassID, FacilityID) → Conditions | PassAccessID, PassID, FacilityID |

## 4.2 Normalization

Normalization was used to remove redundancy caused by the wrong functional dependencies (FDs).

1NF was needed first so each column holds a single value and each row is identified by a key.

2NF removes partial dependencies, where a non-prime attribute depends on only part of a composite candidate key such as (GroupID, FacilityID); every non-prime column here depends on the whole key, so nothing changed.

3NF removes transitive dependencies, where a non-prime attribute depends on another non-prime attribute instead of the key. Two were found, RateID → GroupID in MEMBERSHIP and Term → TermStart, TermEnd in MEMBERSHIP_RATE. So each was fixed by removing the column or moving it to its own table.

**Table 6.** Normalization steps

| Step | Rule checked | Finding | Action |
|---|---|---|---|
| **1NF** | Every column holds one value and there are no repeating groups. | A facility has many opening ranges (one per day). They cannot sit in a single FACILITY column. | Hours stay in OPERATING_HOURS, one row per facility per day. Derived attributes are not stored as columns. |
| **2NF** | No non-prime attribute depends on only part of a composite candidate key. | Every other column depends on the whole key. | No change needed. |
| **3NF (a)** | No non-prime attribute depends on another non-prime attribute. | MEMBERSHIP:<br>MembershipID → RateID and<br>RateID → GroupID<br>So GroupID depends on the key only through RateID. | Remove GroupID from MEMBERSHIP. The group is reached through MEMBERSHIP_RATE. |
| **3NF (b)** | Same rule. | MEMBERSHIP_RATE:<br>RateID → Term and<br>Term → TermStart<br>So the term dates repeat on every rate for that term. | Move Term, TermStart and TermEnd into a new TERM table. MEMBERSHIP_RATE keeps TermID (FK). |

**Table 7.** Tables changed by normalization

| Table | Columns | Functional dependencies | Prime attributes |
|---|---|---|---|
| **MEMBERSHIP** | MembershipID, StartDate, EndDate, PricePaid, RateID (FK) | MembershipID → all other columns | MembershipID |
| **MEMBERSHIP_RATE** | RateID, Price, SpouseAddOn, DependantAddOn, GroupID (FK), DiscountID (FK), TermID (FK) | RateID → all other columns | RateID |
| **TERM (new)** | TermID, TermName, TermStart, TermEnd | TermID → TermName, TermStart, TermEnd | TermID |

*All other tables stay as shown in Table 5. The result is 22 tables, all in 3NF.*

## 5. Outcome

The review leaves a model of 22 entities, Five associative entities (REGISTRATION, MEMBERSHIP_HOLDER, PASS_SERVICE, MEMBERSHIP_FACILITY_ACCESS and PASS_FACILITY_ACCESS) resolves every many-to-many relationship. Each fan trap and chasm trap now has a documented solution, and the fifteen time-variant entities are identified for effective dating.

## Works Cited


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